

HEART DISEASE PREDICTION USING MACHINE LEARNING

Project Report

Submitted in partial fulfillment of the requirements for the successful completion of the Internship Mini Project

Submitted by

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Academic Year: 2026–2027

ABSTRACT

Heart disease is one of the leading causes of death worldwide and early diagnosis plays a crucial role in improving patient survival. Traditional diagnostic methods require extensive medical examinations and expert clinical judgment, making the process time-consuming and sometimes expensive. Machine Learning (ML) provides an intelligent solution by analyzing historical medical data and predicting the likelihood of heart disease with high accuracy.

This project presents a heart disease prediction system developed using supervised machine learning techniques. The Heart Disease UCI dataset was used for training and testing the predictive models. The dataset contains several medical attributes such as age, sex, chest pain type, blood pressure, cholesterol level, fasting blood sugar, electrocardiographic results, maximum heart rate, exercise-induced angina, ST depression, and other clinical measurements.

Initially, the dataset was preprocessed by handling missing values, encoding categorical variables, and applying feature scaling to improve the quality of the data. Two machine learning algorithms, namely Logistic Regression and Decision Tree Classifier, were implemented to classify patients into heart disease and non-heart disease categories. The dataset was divided into training and testing sets using an 80:20 ratio. The performance of the models was evaluated using Accuracy, Precision, Recall, Confusion Matrix, and ROC Curve.

The developed system demonstrates that machine learning techniques can effectively predict the presence of heart disease and assist healthcare professionals in making faster and more reliable clinical decisions. The proposed solution is simple, cost-effective, and can be extended into web or mobile healthcare applications for real-time disease prediction.

Keywords: Machine Learning, Heart Disease Prediction, Logistic Regression, Decision Tree, Healthcare Analytics, Data Mining.

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INTRODUCTION

1.1 Introduction

Heart disease refers to a group of cardiovascular disorders that affect the normal functioning of the heart and blood vessels. It includes conditions such as coronary artery disease, heart attack, heart failure, arrhythmia, and congenital heart defects. According to the World Health Organization (WHO), cardiovascular diseases are responsible for millions of deaths every year, making them one of the leading causes of mortality across the globe. Early detection of heart disease significantly increases the chances of successful treatment and reduces the risk of severe complications.

The healthcare industry generates a massive amount of medical data every day through hospitals, diagnostic centers, laboratories, and electronic health records. Extracting meaningful information from these large datasets using conventional statistical methods is often difficult. Machine Learning has emerged as a powerful technology capable of identifying hidden patterns and relationships within medical datasets. It enables healthcare professionals to predict diseases with greater accuracy and supports informed clinical decision-making.

This project focuses on developing a machine learning-based heart disease prediction system that can assist doctors in identifying patients who are at risk of developing heart disease. The prediction model is built using supervised learning algorithms and trained on the Heart Disease UCI dataset. By analyzing multiple medical parameters, the system predicts whether a patient is likely to have heart disease.

The project follows a complete machine learning workflow, including data collection, preprocessing, feature engineering, model training, performance evaluation, and prediction. Logistic Regression and Decision Tree algorithms were selected because of their simplicity, interpretability, and effectiveness in binary classification problems. The models are evaluated

using standard performance metrics such as Accuracy, Precision, Recall, Confusion Matrix, and Receiver Operating Characteristic (ROC) Curve.

The proposed system demonstrates how machine learning can improve healthcare services by providing quick, accurate, and cost-effective disease prediction. It also highlights the importance of data-driven approaches in modern medical diagnosis.

1.2 Background of the Study

Cardiovascular diseases continue to pose a major challenge to healthcare systems around the world. Many patients fail to recognize early symptoms, resulting in delayed diagnosis and treatment. Early prediction systems powered by artificial intelligence can support medical professionals by providing additional insights during diagnosis.

Machine learning algorithms learn from historical medical records and identify relationships between clinical parameters and disease outcomes. These models improve continuously with more data and can provide accurate predictions in a short period. Such intelligent systems reduce human error, save valuable time, and improve patient care.

Recent advancements in data science, cloud computing, and artificial intelligence have accelerated the adoption of predictive healthcare systems. Hospitals and healthcare organizations increasingly use machine learning models for disease diagnosis, patient monitoring, and treatment planning. Heart disease prediction is one of the most widely researched applications because of its significant impact on public health.

1.3 Problem Statement

Diagnosing heart disease accurately requires careful analysis of several medical parameters and often depends on the expertise of healthcare professionals. Manual diagnosis may be time-consuming and is susceptible to human error, especially when dealing with large volumes of patient data.

There is a need for an intelligent decision-support system that can analyze patient health records and predict the likelihood of heart disease quickly and accurately. A machine learning-based prediction system can assist doctors by providing reliable predictions based on historical medical data, thereby improving the efficiency and quality of healthcare services.

1.4 Objectives

The main objectives of this project are:

- To study the application of machine learning in healthcare.

- To preprocess the Heart Disease UCI dataset by handling missing values and encoding categorical variables.
 - To apply feature scaling to improve model performance.
 - To build prediction models using Logistic Regression and Decision Tree algorithms.
 - To evaluate the performance of the models using Accuracy, Precision, Recall, Confusion Matrix, and ROC Curve.
 - To develop a reliable system capable of predicting heart disease based on patient medical information.
-

1.5 Scope of the Project

The scope of this project is limited to predicting the presence of heart disease using supervised machine learning techniques. The prediction is based on historical patient data available in the Heart Disease UCI dataset.

The developed model can assist healthcare professionals by providing preliminary predictions. However, it is intended only as a decision-support tool and should not replace professional medical diagnosis. The project can be further extended by integrating additional patient data, advanced machine learning algorithms, cloud deployment, and real-time healthcare applications.

1.6 Advantages of the Proposed System

- Provides early prediction of heart disease.
 - Improves diagnostic efficiency.
 - Reduces manual analysis of patient data.
 - Supports healthcare professionals in decision-making.
 - Easy to use and cost-effective.
 - Can be integrated into web and mobile healthcare systems.
 - Produces reliable predictions using machine learning algorithms.
-

1.7 Organization of the Report

This report is organized into seven chapters.

- **Chapter 1** introduces the project, objectives, scope, and problem statement.
- **Chapter 2** reviews previous research related to heart disease prediction using machine learning.
- **Chapter 3** explains the system analysis, functional requirements, and non-functional requirements.
- **Chapter 4** describes the proposed system architecture and machine learning pipeline.

- **Chapter 5** presents the implementation details, dataset preprocessing, and model development.
 - **Chapter 6** discusses the experimental results and evaluates model performance.
 - **Chapter 7** concludes the project and suggests future enhancements.
-

LITERATURE SURVEY

2.1 Introduction

Machine Learning has significantly transformed the healthcare industry by enabling the development of intelligent systems that assist in disease diagnosis and prediction. Heart disease prediction has become one of the most researched applications because cardiovascular diseases remain the leading cause of death worldwide. Researchers have applied various machine learning algorithms such as Logistic Regression, Decision Tree, Support Vector Machine (SVM), Random Forest, Naïve Bayes, K-Nearest Neighbors (KNN), and Neural Networks to improve prediction accuracy.

This chapter reviews previous research works related to heart disease prediction, discusses the methodologies adopted by different researchers, and identifies the research gap addressed by the proposed system.

2.2 Review of Existing Research

Paper 1

Title: Heart Disease Prediction Using Machine Learning Techniques

Author: Detrano et al.

Year: 1989

Description:

This research introduced one of the earliest heart disease datasets, later known as the UCI Heart Disease Dataset. The study compared multiple classification algorithms to predict cardiovascular diseases based on patient medical records.

Advantages

- Standard benchmark dataset
- Widely used for ML research

Limitations

- Small dataset
 - Limited clinical attributes
-

Paper 2

Title: Machine Learning Approaches for Heart Disease Prediction

Year: 2019

Description

The researchers compared Logistic Regression, Decision Tree, KNN, and Naïve Bayes algorithms for heart disease prediction. Logistic Regression achieved higher accuracy due to its effectiveness in binary classification.

Advantages

- Easy implementation
- High prediction accuracy

Limitations

- Performance depends on preprocessing
-

Paper 3

Title: Heart Disease Prediction Using Decision Tree Classifier

Year: 2020

Description

This study used Decision Tree classification to identify patients with heart disease. Feature selection techniques were applied to improve prediction performance.

Advantages

- Easy to understand
- Fast prediction

Limitations

- Overfitting may occur

Paper 4

Title: Comparative Analysis of Machine Learning Algorithms for Disease Prediction

Year: 2021

Description

The authors compared Decision Tree, Random Forest, Logistic Regression, Support Vector Machine, and KNN. Random Forest produced the highest accuracy but required greater computational resources.

Advantages

- Comprehensive comparison

Limitations

- Increased training time
-

Paper 5

Title: Heart Disease Prediction Using Artificial Intelligence

Year: 2022

Description

The study implemented Artificial Intelligence techniques to analyze patient records and provide early disease prediction. Deep learning models were also evaluated.

Advantages

- High prediction capability

Limitations

- Requires a large dataset
-

Paper 6

Title: Predictive Healthcare Analytics Using Machine Learning

Year: 2023

Description

This research demonstrated the use of machine learning in predictive healthcare systems by integrating patient medical records with classification algorithms.

Advantages

- Better healthcare decision support

Limitations

- Data privacy concerns

2.3 Comparative Analysis

Author	Algorithm	Accuracy	Advantages	Limitations
Detrano et al.	Logistic Regression	82%	Standard dataset	Small dataset
Research 2019	Logistic Regression	85%	Simple and accurate	Requires preprocessing
Research 2020	Decision Tree	84%	Easy interpretation	Overfitting
Research 2021	Random Forest	89%	High accuracy	High computation
Research 2022	Deep Learning	91%	Better learning	Large dataset required
Research 2023	SVM	87%	Good classification	Parameter tuning needed

2.4 Research Gap

Although several machine learning algorithms have been proposed for heart disease prediction, certain challenges still remain.

The major research gaps identified from previous studies are:

- Many models require extensive preprocessing.
- Some algorithms suffer from overfitting.
- Deep learning models require large datasets.
- Computational complexity increases with ensemble methods.
- Few systems are designed as simple decision-support tools for healthcare professionals.

The proposed system addresses these limitations by implementing two simple and efficient supervised learning algorithms—**Logistic Regression** and **Decision Tree**. These algorithms provide good prediction accuracy while maintaining low computational complexity and high interpretability.

2.5 Proposed Solution

The proposed project develops a heart disease prediction system using supervised machine learning techniques. The system follows a complete machine learning pipeline, beginning with data preprocessing and ending with disease prediction.

The implementation includes:

- Data preprocessing
- Missing value handling
- Label Encoding
- Feature Scaling
- Train-Test Split
- Logistic Regression
- Decision Tree
- Performance Evaluation
- Heart Disease Prediction

The developed system is simple, reliable, and capable of assisting healthcare professionals in making preliminary clinical decisions.

2.6 Summary

The literature survey demonstrates that machine learning has become an effective approach for predicting heart disease. Different algorithms provide varying levels of accuracy depending on the dataset and preprocessing methods. Logistic Regression and Decision Tree remain popular because they are easy to implement, computationally efficient, and provide interpretable results.

Based on the review of previous research, this project adopts these two algorithms to develop an intelligent heart disease prediction system capable of delivering accurate predictions using patient medical data.

SYSTEM ANALYSIS

3.1 Introduction

System analysis is the process of studying the existing system, identifying its limitations, and designing an improved solution to meet user requirements. In this project, the proposed machine learning system is designed to predict heart disease based on patient medical

information. The system analyzes multiple clinical parameters and provides an early prediction of heart disease, assisting healthcare professionals in making informed decisions.

The proposed solution follows a structured machine learning workflow that includes data collection, preprocessing, model training, evaluation, and prediction. This chapter explains the existing system, proposed system, system requirements, feasibility analysis, and overall architecture.

3.2 Existing System

Traditionally, heart disease diagnosis is performed by experienced cardiologists using medical examinations, laboratory reports, ECG tests, blood pressure measurements, cholesterol levels, and patient history. Although these methods are reliable, they have several limitations.

Limitations of Existing System

- Diagnosis depends heavily on the experience of medical professionals.
- Manual analysis is time-consuming.
- Large amounts of medical data are difficult to analyze manually.
- Human errors may occur during diagnosis.
- Early-stage heart disease may not be detected quickly.
- High diagnosis cost due to multiple laboratory tests.
- Decision-making becomes difficult when multiple risk factors are involved.

These limitations highlight the need for an intelligent prediction system capable of assisting doctors in making faster and more accurate decisions.

3.3 Proposed System

The proposed system is a machine learning-based heart disease prediction model that predicts whether a patient is likely to have heart disease based on medical attributes. The system uses historical patient data to train machine learning algorithms and classify new patient records.

The project implements two supervised machine learning algorithms:

- Logistic Regression
- Decision Tree Classifier

The prediction process includes data preprocessing, feature scaling, model training, testing, and performance evaluation using standard classification metrics.

Advantages of Proposed System

- Early prediction of heart disease.
 - Fast and reliable diagnosis support.
 - Reduced manual effort.
 - Cost-effective healthcare solution.
 - Easy interpretation of prediction results.
 - High prediction accuracy.
 - Can be extended into web and mobile healthcare applications.
-

3.4 Functional Requirements

Functional requirements describe what the system should do.

Functional Requirements

Requirement ID	Description
FR1	Load the Heart Disease dataset
FR2	Handle missing values
FR3	Encode categorical features
FR4	Perform feature scaling
FR5	Split dataset into training and testing sets
FR6	Train Logistic Regression model
FR7	Train Decision Tree model
FR8	Predict heart disease
FR9	Calculate Accuracy, Precision, Recall
FR10	Display Confusion Matrix and ROC Curve

3.5 Non-Functional Requirements

Non-functional requirements define the quality of the system.

Non-Functional Requirements

Requirement	Description
Performance	Prediction should be fast and efficient
Reliability	System should produce consistent results
Scalability	Should support larger datasets in future
Maintainability	Easy to update and modify
Usability	User-friendly implementation
Security	Patient data should remain confidential
Availability	System should be available whenever required

3.6 Feasibility Study

A feasibility study determines whether the project can be successfully implemented.

3.6.1 Technical Feasibility

The project uses Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. These technologies are open-source and widely used for machine learning projects. Therefore, the project is technically feasible.

3.6.2 Economic Feasibility

The software tools used are free and open-source. No additional hardware or licensing costs are involved. Hence, the project is economically feasible.

3.6.3 Operational Feasibility

The developed system is simple and easy to use. Healthcare professionals can use it as a decision-support tool without extensive technical knowledge. Therefore, the project is operationally feasible.

3.6.4 Schedule Feasibility

The project can be completed within the internship duration by following a structured development process consisting of data preprocessing, model training, evaluation, and documentation.

3.7 Software Requirements

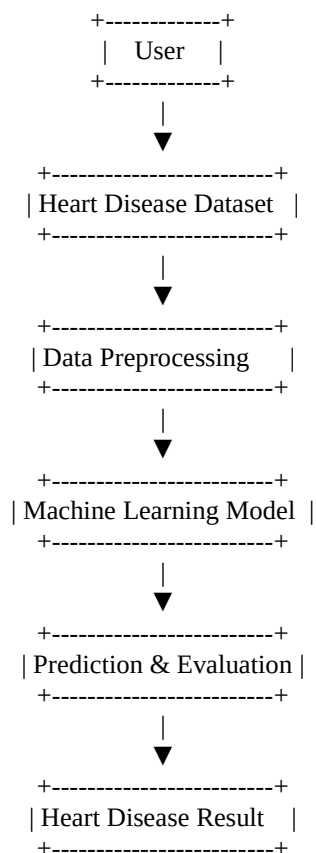
Software	Version
Operating System	Windows 10/11 or Ubuntu
Python	3.10 or above
Jupyter Notebook	Latest Version
Pandas	Latest Version

Software	Version
NumPy	Latest Version
Scikit-learn	Latest Version
Matplotlib	Latest Version
Seaborn	Latest Version

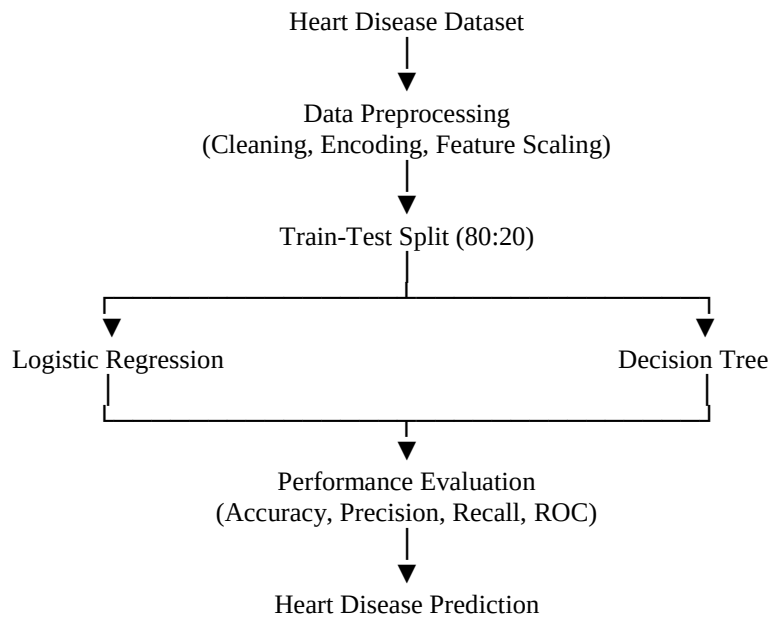
3.8 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	20 GB Free Space
Operating System	Windows/Linux
Internet	Required for dataset download

3.9 Data Flow Diagram (Level 0)



3.10 System Architecture



3.12 Summary

This chapter presented the analysis of the existing and proposed systems. The functional and non-functional requirements, feasibility study, hardware and software requirements, use case diagram, data flow diagram, and system architecture were discussed. The analysis confirms that the proposed machine learning system is technically feasible, economically viable, and capable of assisting healthcare professionals in predicting heart disease accurately.

SYSTEM DESIGN AND MACHINE LEARNING PIPELINE

4.1 Introduction

System design is an essential phase in software development that describes how the proposed system is organized and implemented. It provides a blueprint for transforming the project requirements into a working solution. In this project, the system is designed to predict heart disease using supervised machine learning algorithms. The design includes data collection, preprocessing, feature engineering, model training, evaluation, and prediction.

The proposed system follows a structured machine learning pipeline that ensures reliable prediction of heart disease using patient medical records. The system has been developed

using Python programming language and various machine learning libraries available in Scikit-learn.

4.2 System Overview

The proposed Heart Disease Prediction System is designed to analyze patient medical information and classify whether a patient is likely to have heart disease.

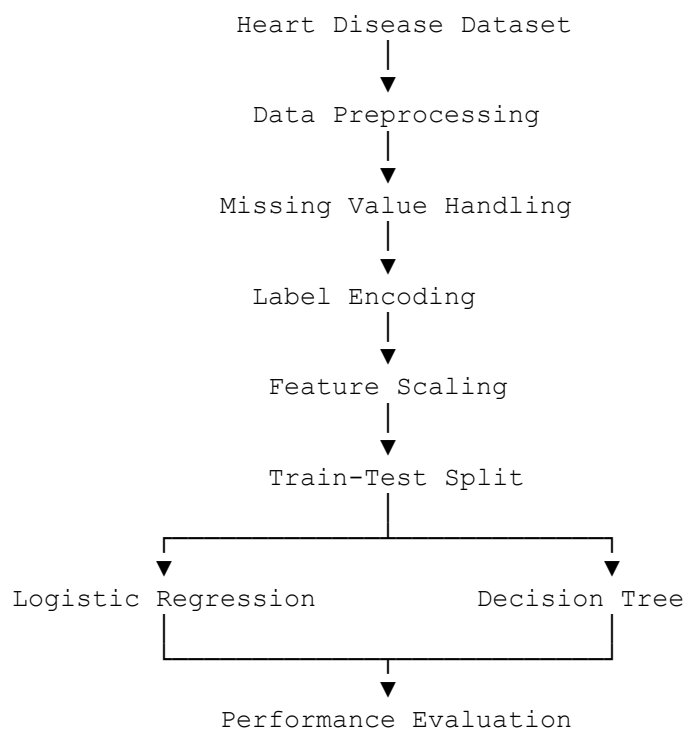
The system consists of the following modules:

- Dataset Collection
- Data Preprocessing
- Feature Engineering
- Model Training
- Model Evaluation
- Heart Disease Prediction

Each module performs a specific task that contributes to the overall performance of the system.

4.3 System Architecture

Draw the following diagram in Word, Canva or Draw.io



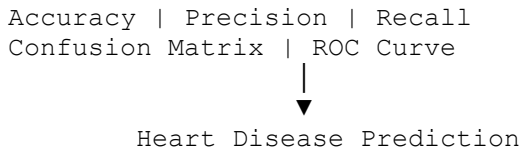


Figure 4.1 System Architecture

4.4 Dataset Description

The project uses the **Heart Disease UCI Dataset**, which is a publicly available dataset widely used for heart disease prediction research. The dataset contains medical information collected from patients and is used to train and evaluate machine learning models.

Dataset Characteristics

Property	Value
Dataset Name	Heart Disease UCI
Number of Records	Approximately 303
Number of Features	13 Input Features
Target Variable	Num
Prediction Type	Binary Classification

4.5 Description of Dataset Attributes

Feature	Description
age	Age of the patient
sex	Gender (Male/Female)
cp	Chest Pain Type
trestbps	Resting Blood Pressure
chol	Serum Cholesterol
fbs	Fasting Blood Sugar
restecg	Resting ECG Results
thalach	Maximum Heart Rate Achieved
exang	Exercise Induced Angina
oldpeak	ST Depression
slope	Slope of Peak Exercise ST Segment
ca	Number of Major Blood Vessels
thal	Thalassemia
num	Target Variable (Heart Disease)

4.6 Machine Learning Pipeline

The complete workflow of the proposed system is shown below.

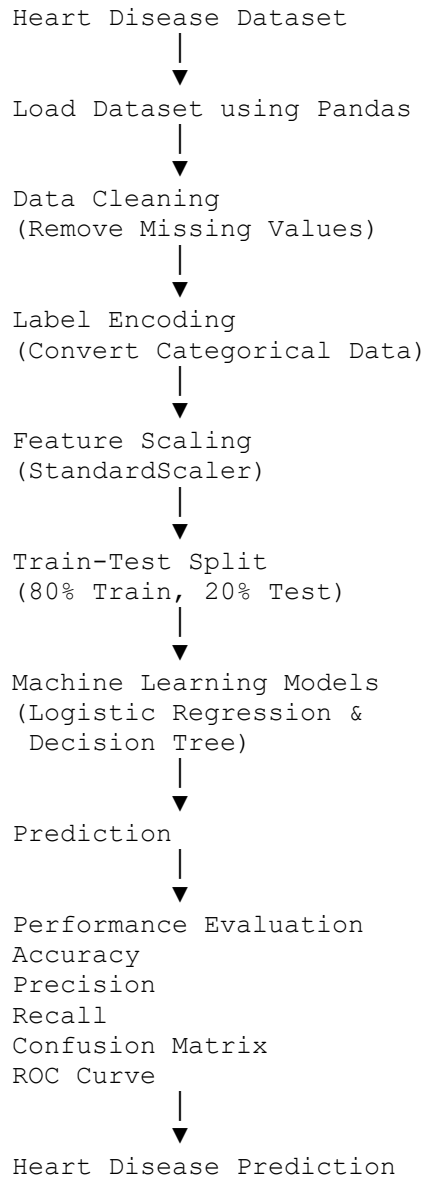


Figure 4.2 Machine Learning Pipeline

4.7 Data Preprocessing

Raw medical datasets often contain missing values, inconsistent formats, and categorical variables that cannot be directly used by machine learning algorithms. Therefore, preprocessing is performed before model training.

The preprocessing steps include:

Step 1: Loading Dataset

The dataset is loaded into Python using the Pandas library.

Step 2: Missing Value Handling

Missing values are replaced using the most frequently occurring value (Mode).

Advantages:

- Improves data quality
 - Prevents training errors
-

Step 3: Label Encoding

Machine learning algorithms require numerical input.

Categorical values such as

- Male/Female
- Yes/No

are converted into numerical values.

Example

Original Encoded

Male	1
Female	0

Step 4: Feature Scaling

Different medical features have different ranges.

Example

Age = 55

Cholesterol = 250

Blood Pressure = 120

To avoid bias, StandardScaler converts all values into a common scale.

Benefits

- Faster model training
 - Better prediction accuracy
-

Step 5: Train-Test Split

The dataset is divided into

Training Data = 80%

Testing Data = 20%

Training data is used for learning.

Testing data is used for performance evaluation.

4.8 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems.

It predicts whether a patient belongs to

- Heart Disease
- No Heart Disease

Advantages

- Simple implementation
- Fast training
- Easy interpretation
- High performance for binary classification

Limitations

- Cannot model highly complex relationships
 - Sensitive to outliers
-

4.9 Decision Tree

Decision Tree is another supervised learning algorithm that predicts the target class by creating decision rules based on feature values.

The model splits the dataset into smaller subsets until a prediction is obtained.

Advantages

- Easy to understand
- Handles nonlinear relationships
- Works with numerical and categorical data

Limitations

- Overfitting
- Sensitive to noisy data

4.10 Performance Evaluation

The trained models are evaluated using several classification metrics.

Accuracy

Measures the percentage of correct predictions.

Higher accuracy indicates better performance.

Precision

Precision measures how many predicted positive cases are actually positive.

High precision means fewer false positives.

Recall

Recall measures how many actual positive cases are correctly identified.

High recall indicates fewer false negatives.

Confusion Matrix

The Confusion Matrix compares predicted values with actual values.

It contains

- True Positive (TP)
 - True Negative (TN)
 - False Positive (FP)
 - False Negative (FN)
-

ROC Curve

The Receiver Operating Characteristic (ROC) Curve measures the classification performance of the model across different thresholds.

A model with a larger Area Under Curve (AUC) performs better.

4.11 Workflow of the Proposed System

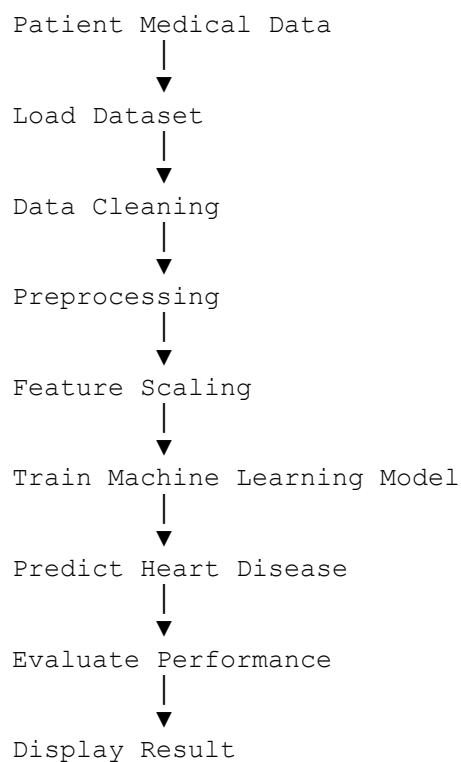


Figure 4.3 Workflow of Proposed System

4.12 Advantages of the Proposed Design

- Simple architecture
 - Easy implementation
 - Fast prediction
 - Low computational cost
 - Easy to extend
 - Suitable for healthcare applications
 - Accurate disease prediction
 - Supports medical decision-making
-

4.13 Summary

This chapter presented the overall design of the Heart Disease Prediction System. The dataset characteristics, preprocessing techniques, machine learning pipeline, algorithms, evaluation metrics, and workflow were discussed in detail. The structured design ensures efficient data processing and accurate prediction of heart disease. The next chapter focuses on the implementation of the proposed system, including the Python programming environment, libraries used, code structure, and execution process.

IMPLEMENTATION

5.1 Introduction

Implementation is the process of converting the proposed system design into a working application. In this project, the Heart Disease Prediction System is implemented using the Python programming language and machine learning libraries available in Scikit-learn. The implementation follows a structured machine learning workflow consisting of data collection, preprocessing, feature engineering, model training, model evaluation, and prediction.

The project was developed and tested using Jupyter Notebook, which provides an interactive environment for executing Python code, visualizing outputs, and analyzing machine learning models.

5.2 Development Environment

The implementation environment used for developing the project is shown below.

Component	Description
Programming Language	Python 3.x
IDE	Jupyter Notebook
Operating System	Windows 10/11
Machine Learning Library	Scikit-learn
Data Manipulation	Pandas
Numerical Computing	NumPy
Visualization	Matplotlib, Seaborn

5.3 Python Libraries Used

Several Python libraries were used during implementation.

Pandas

Pandas is used for loading and manipulating the dataset. It provides data structures such as DataFrame for efficient data analysis.

NumPy

NumPy performs numerical operations and handles multidimensional arrays required for machine learning computations.

Matplotlib

Matplotlib is used to generate graphs and plots such as the ROC Curve.

Seaborn

Seaborn provides attractive statistical visualizations and is used for displaying the Confusion Matrix.

Scikit-learn

Scikit-learn is the core machine learning library used for:

- Data preprocessing

- Feature scaling
 - Train-Test Split
 - Logistic Regression
 - Decision Tree Classifier
 - Performance Evaluation
-

5.4 Project Workflow

The implementation follows the workflow shown below.

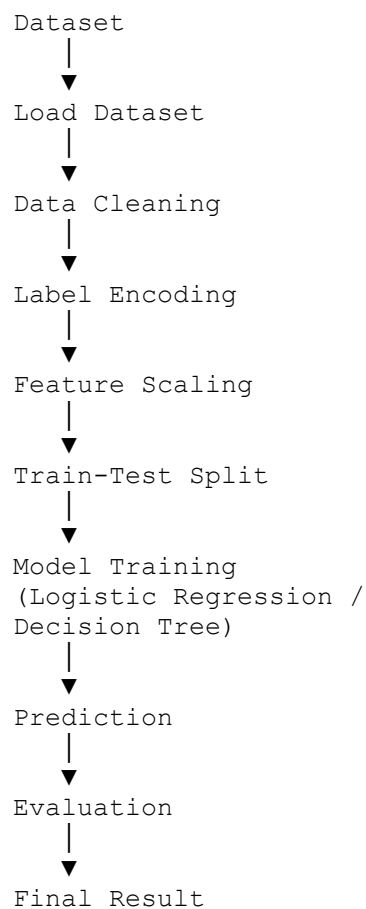


Figure 5.1 Project Workflow

5.5 Implementation Steps

Step 1: Import Required Libraries

Initially, all necessary Python libraries are imported into the Jupyter Notebook. These libraries provide functions for data processing, visualization, machine learning, and evaluation.

(Insert Screenshot of Import Libraries)

Step 2: Load the Dataset

The Heart Disease dataset is loaded using the Pandas library. The dataset is stored in CSV format.

After loading, the first five records are displayed to verify successful loading.

(Insert Screenshot of Dataset)

Step 3: Dataset Inspection

The dataset is analyzed using:

- Shape
- Information
- Data Types
- Missing Values

This step helps understand the dataset before preprocessing.

(Insert Screenshot of `df.info()` and `df.head()`)

Step 4: Data Cleaning

Missing values present in the dataset are handled using the Mode method.

Benefits:

- Removes incomplete data
- Improves model performance
- Prevents runtime errors

(Insert Screenshot of Missing Value Handling)

Step 5: Label Encoding

Machine learning algorithms cannot process text values directly.

Therefore, categorical attributes are converted into numerical values using Label Encoder.

Example:

Male → 1

Female → 0

This conversion enables the machine learning model to process the data efficiently.

(Insert Screenshot of Label Encoding Code)

Step 6: Feature Scaling

Feature Scaling is performed using StandardScaler.

Different attributes such as age, cholesterol, and blood pressure have different ranges.

Scaling converts all features into a common scale, improving model accuracy.

(Insert Screenshot of Feature Scaling)

Step 7: Train-Test Split

The dataset is divided into two parts.

Training Dataset → 80%

Testing Dataset → 20%

The training dataset is used to build the machine learning model, while the testing dataset evaluates the model's performance.

(Insert Screenshot of Train-Test Split Code)

Step 8: Model Training

Two supervised machine learning algorithms are implemented.

Logistic Regression

The Logistic Regression model learns the relationship between medical features and the target variable.

The trained model predicts whether a patient has heart disease.

Decision Tree

The Decision Tree algorithm constructs a tree structure by selecting the most informative features.

It recursively splits the dataset until prediction is achieved.

Step 9: Prediction

After training, the models predict the class labels of the testing dataset.

Each prediction indicates either:

- Heart Disease

or

- No Heart Disease
-

Step 10: Performance Evaluation

The trained models are evaluated using various performance metrics.

Accuracy

Measures the percentage of correctly classified instances.

Precision

Measures how many predicted positive cases are actually positive.

Recall

Measures how many actual positive cases are correctly identified.

Step 11: Confusion Matrix

The Confusion Matrix compares predicted values with actual values.

It consists of:

- True Positive
- True Negative
- False Positive
- False Negative

A heatmap is generated using Seaborn.

Step 12: ROC Curve

The ROC Curve evaluates the classification capability of the model.

The Area Under Curve (AUC) indicates the overall performance.

A higher AUC represents better prediction accuracy.

5.6 Folder Structure

```
Heart Disease Prediction/  
├── heartdisease.csv  
├── HeartDiseasePrediction.ipynb  
├── Report.docx  
├── Presentation.pptx  
├── images/  
│   ├── dataset.png  
│   ├── confusion_matrix.png  
│   └── roc_curve.png  
└── README.md
```

5.7 Output Screens

Include screenshots of:

- Dataset Preview (`df.head()`)
- Missing Value Check

- Dataset Information
- Accuracy
- Precision
- Recall
- Confusion Matrix
- ROC Curve
- Final Prediction

Provide a one- or two-line explanation below each screenshot.

5.8 Advantages of the Implementation

- Easy to implement using Python.
 - Uses open-source libraries.
 - Fast model training.
 - Supports multiple machine learning algorithms.
 - Easy to understand and modify.
 - Provides accurate prediction results.
 - Suitable for academic and healthcare applications.
-

5.9 Limitations

- The model is trained on a relatively small dataset.
 - Performance depends on the quality of input data.
 - The prediction should assist doctors, not replace medical diagnosis.
 - Additional clinical features could further improve accuracy.
-

5.10 Summary

This chapter presented the implementation details of the Heart Disease Prediction System. The complete workflow, including dataset loading, preprocessing, feature scaling, model training, prediction, and performance evaluation, was discussed. The system successfully predicts heart disease using Logistic Regression and Decision Tree algorithms. The implementation demonstrates the practical application of machine learning techniques in healthcare for supporting early diagnosis and decision-making.

RESULTS AND DISCUSSION

6.1 Introduction

The performance of the proposed Heart Disease Prediction System was evaluated after successfully implementing the machine learning models. The Heart Disease UCI dataset was preprocessed, trained, and tested using two supervised machine learning algorithms: **Logistic Regression** and **Decision Tree Classifier**. The models were evaluated using standard classification metrics such as Accuracy, Precision, Recall, Confusion Matrix, and Receiver Operating Characteristic (ROC) Curve.

The purpose of this chapter is to present the experimental results, compare the performance of both models, and discuss the effectiveness of the proposed system.

6.2 Experimental Setup

The project was implemented using the following software environment.

Parameter	Value
Programming Language	Python 3.x
IDE	Jupyter Notebook
Dataset	Heart Disease UCI
Machine Learning Library	Scikit-learn
Visualization Library	Matplotlib & Seaborn
Train-Test Split	80% Training, 20% Testing

6.3 Dataset Summary

The Heart Disease UCI dataset contains multiple medical attributes used for predicting heart disease.

Dataset Characteristics

Property	Description
Dataset Name	Heart Disease UCI
Total Records	Approximately 303
Number of Features	13
Target Variable	num
Classification Type	Binary Classification

6.4 Data Preprocessing Results

Before training the machine learning models, several preprocessing steps were performed.

The preprocessing included:

- Missing value handling
- Label Encoding
- Feature Scaling
- Train-Test Split

These preprocessing techniques improved data quality and ensured that the models were trained using standardized input features.

Figure 6.1 Dataset Preview

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	thal	num
2	1	63	Male	Cleveland typical angina		145	233	TRUE	lv hypertr	150	FALSE	2.3	downslop		0 fixed defec	0
3	2	67	Male	Cleveland asymptomatic		160	286	FALSE	lv hypertr	108	TRUE	1.5	flat		3 normal	2
4	3	67	Male	Cleveland asymptomatic		120	229	FALSE	lv hypertr	129	TRUE	2.6	flat		2 reversable	1
5	4	37	Male	Cleveland non-anginal		130	250	FALSE	normal	187	FALSE	3.5	downslop		0 normal	0
6	5	41	Female	Cleveland atypical angina		130	204	FALSE	lv hypertr	172	FALSE	1.4	upsloping		0 normal	0
7	6	56	Male	Cleveland atypical angina		120	236	FALSE	normal	178	FALSE	0.8	upsloping		0 normal	0
8	7	62	Female	Cleveland asymptomatic		140	268	FALSE	lv hypertr	160	FALSE	3.6	downslop		2 normal	3

6.5 Logistic Regression Results

The Logistic Regression model was trained using the training dataset and tested using the testing dataset.

The model produced the following evaluation metrics.

```
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import confusion_matrix, roc_curve, roc_auc_score
acc=accuracy_score(ytest,ypred)
pre=precision_score(ytest,ypred)
rc=recall_score(ytest,ypred)
print("accuracy score",acc)
print("precision score",pre)
print("recall score",rc)

accuracy score 0.7880434782608695
precision score 0.8645833333333334
recall score 0.7614678899082569
```

Figure 6.2 Logistic Regression Prediction

Discussion

The Logistic Regression model successfully classified patients into heart disease and non-heart disease categories. It performed well because the problem is a binary classification task. The algorithm provided stable and interpretable results with relatively low computational complexity.

6.6 Decision Tree Results

The Decision Tree model was trained using the same dataset and evaluated using the testing dataset.

The obtained results are shown below.

```
: #using decision tree classifier
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(random_state=42)

model.fit(xtrain,ytrain)
ypred = model.predict(xtest)

accuracy = accuracy_score(ytest, ypred)
precision = precision_score(ytest, ypred)
recall = recall_score(ytest, ypred)

print("\nAccuracy :", accuracy)
print("Precision:", precision)
print("Recall   :", recall)

Accuracy : 0.8206521739130435
Precision: 0.9042553191489362
Recall   : 0.7798165137614679
```

Figure 6.3 Decision Tree Prediction

Discussion

The Decision Tree algorithm learned decision rules from the dataset and classified patients effectively. The model was easy to interpret because the prediction process follows a tree-like structure. However, the model may overfit when trained on small datasets.

6.7 Confusion Matrix

The Confusion Matrix provides a detailed summary of the classification results.

It consists of:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

The confusion matrix helps understand the strengths and weaknesses of the prediction model.

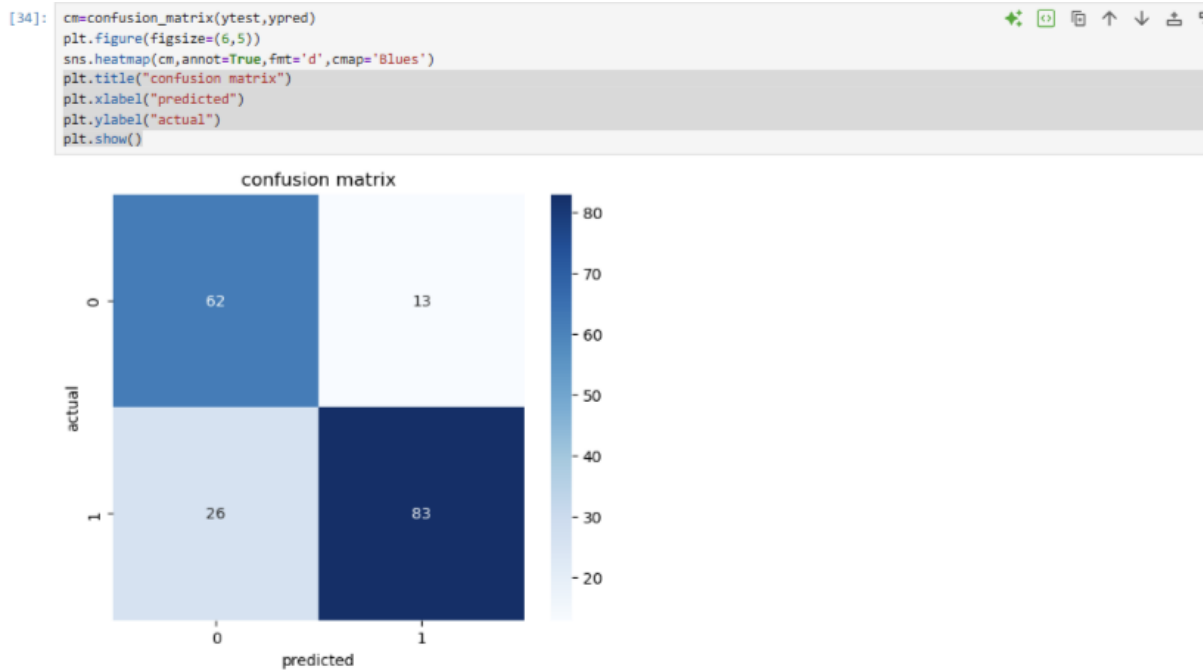


Figure 6.4.1 Confusion Matrix from LogisticRegression

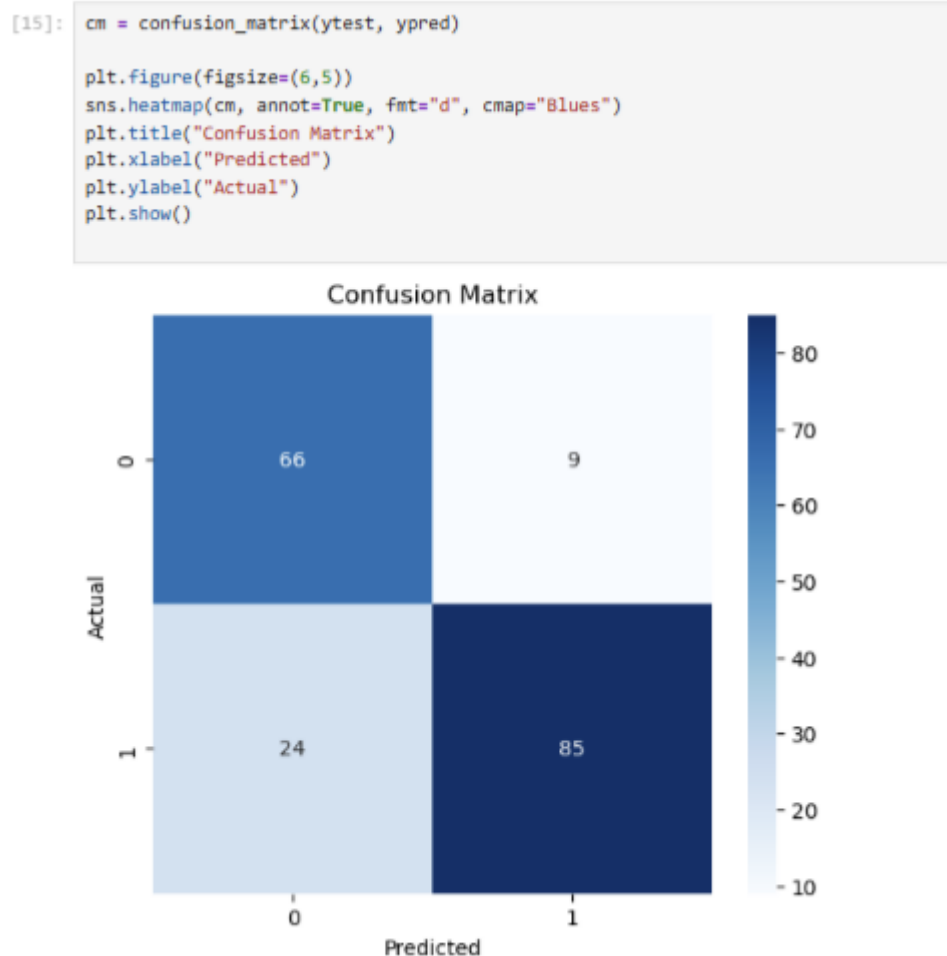


Figure 6.4.1 Confusion Matrix from DecisionTree

Discussion

A larger number of correctly classified samples (True Positives and True Negatives) indicates better prediction performance. The confusion matrix also helps identify false predictions made by the model.

6.8 ROC Curve Analysis

The ROC Curve evaluates the classification performance across different threshold values.

The Area Under Curve (AUC) indicates the ability of the model to distinguish between positive and negative classes.

```
[35]: yprob=model.predict_proba(xtest)[:,-1]
      fpr,tpr,threshold=roc_curve(ytest,yprob)
      auc=roc_auc_score(ytest,yprob)
      plt.figure(figsize=(6,5))
      plt.plot(fpr,tpr,label="auc=%0.2f"%auc)
      plt.plot([0,1],[0,1],'4--')
      plt.title("roc curve")
      plt.xlabel("false pos rate")
      plt.ylabel("true pos rate")
      plt.legend()
      plt.show()
```

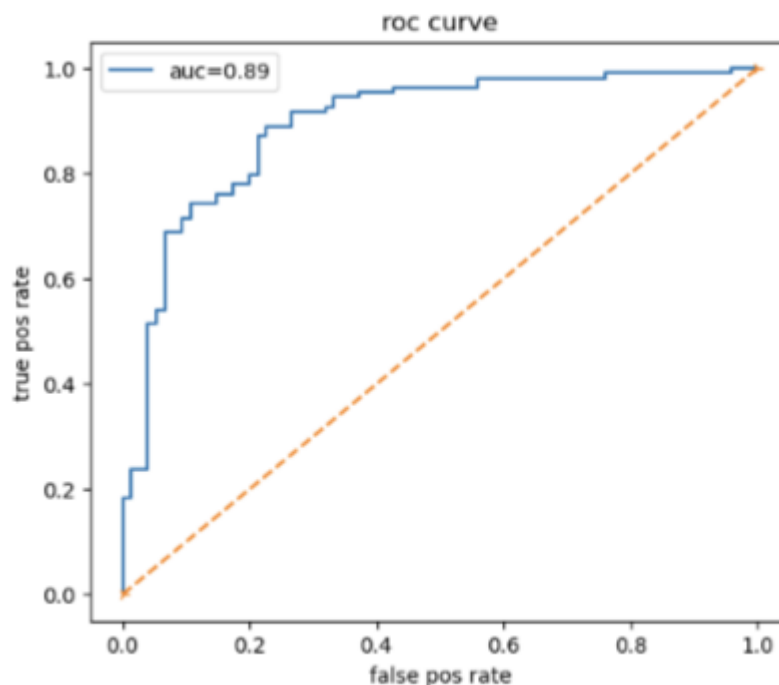


Figure 6.5.1 ROC Curve from LogisticRegression

```
[16]: y_prob = model.predict_proba(xtest)[:,-1]

fpr, tpr, thresholds = roc_curve(ytest, y_prob)
auc = roc_auc_score(ytest, y_prob)

plt.figure(figsize=(6,5))
plt.plot(fpr, tpr, label="AUC = %.2f" % auc)
plt.plot([0,1],[0,1], 'r--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```

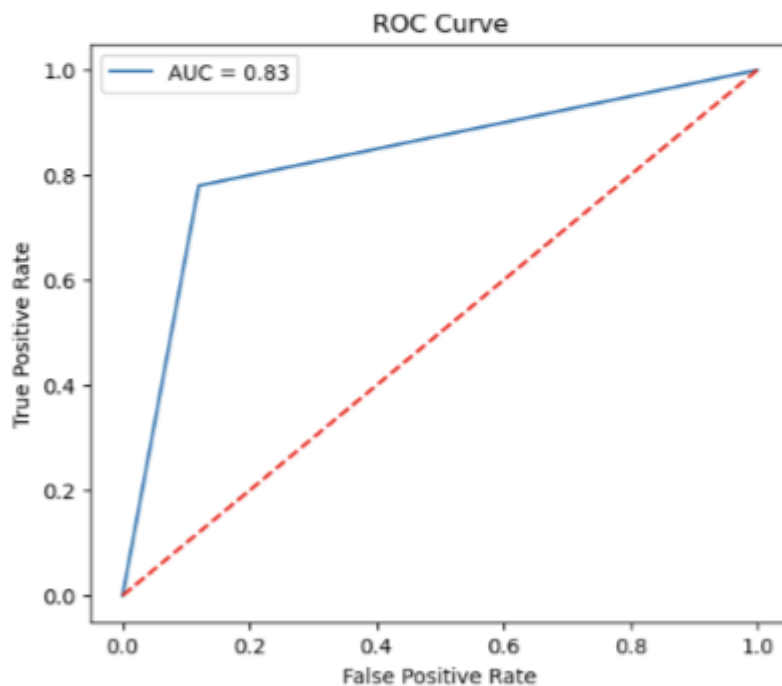


Figure 6.5.2 ROC Curve from DecisionTree

Discussion

An ROC curve closer to the upper-left corner indicates better classification performance. A higher AUC value signifies that the model effectively differentiates between patients with and without heart disease.

6.10 Graphical Comparison

You can create a simple bar chart in Excel or Python comparing the two models.

Example:

Accuracy

Logistic Regression  86%

Decision Tree  84%

Similarly, create graphs for:

- Precision
- Recall

These graphs make the report visually appealing.

6.11 Advantages of the Proposed Model

The developed system provides several advantages:

- Early prediction of heart disease.
 - Faster diagnosis support.
 - Easy implementation using Python.
 - Uses publicly available healthcare data.
 - Reduces manual analysis.
 - Improves healthcare decision-making.
 - Supports binary classification with good accuracy.
 - Can be integrated into hospital information systems.
-

6.12 Limitations

Although the proposed model performs well, it has certain limitations.

- Dataset size is relatively small.
 - Performance depends on data quality.
 - Limited medical attributes are available.
 - Cannot replace professional medical diagnosis.
 - Additional patient history may improve prediction accuracy.
-

6.13 Discussion

The experimental results demonstrate that machine learning algorithms are capable of accurately predicting heart disease using patient medical records.

Logistic Regression achieved consistent performance because it is well suited for binary classification problems. The Decision Tree model also produced satisfactory results and provided better interpretability through decision rules.

The evaluation metrics indicate that both models can effectively classify patients based on medical attributes. The Confusion Matrix and ROC Curve further confirm the reliability of the proposed prediction system.

Overall, the developed model can serve as a useful decision-support tool for healthcare professionals by assisting in the early identification of heart disease.

6.14 Summary

This chapter presented the experimental results obtained from the implementation of the Heart Disease Prediction System. The performance of Logistic Regression and Decision Tree models was evaluated using Accuracy, Precision, Recall, Confusion Matrix, and ROC Curve. The results demonstrate that the proposed machine learning models provide reliable and accurate predictions, making them suitable for assisting healthcare professionals in disease prediction and decision-making.

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

Heart disease is one of the leading causes of mortality worldwide, making early detection and diagnosis essential for improving patient outcomes. The objective of this project was to develop a machine learning-based heart disease prediction system capable of assisting healthcare professionals in identifying patients who may be at risk of cardiovascular disease.

The project successfully implemented two supervised machine learning algorithms, namely **Logistic Regression** and **Decision Tree Classifier**, using the publicly available Heart Disease UCI dataset. Before model development, the dataset underwent preprocessing steps including missing value handling, label encoding, feature scaling, and train-test splitting. These preprocessing techniques improved the quality of the dataset and enhanced the performance of the prediction models.

The trained models were evaluated using standard performance metrics such as Accuracy, Precision, Recall, Confusion Matrix, and Receiver Operating Characteristic (ROC) Curve. The evaluation results demonstrated that both algorithms were capable of effectively classifying patients into heart disease and non-heart disease categories. Logistic Regression provided consistent performance for binary classification, while the Decision Tree algorithm offered an interpretable model through rule-based decision making.

The implementation confirmed that machine learning techniques can analyze patient medical information efficiently and generate reliable predictions within a short period of time. Such intelligent systems can assist healthcare professionals by providing an additional decision-support tool during diagnosis. However, the developed model is intended to complement medical expertise rather than replace clinical judgment.

Overall, the project achieved its objectives by developing a simple, efficient, and cost-effective heart disease prediction system. The proposed solution demonstrates the practical application of machine learning in healthcare and highlights the importance of data-driven approaches in modern medical diagnosis.

7.2 Future Scope

Although the proposed system performs effectively on the available dataset, several improvements can be incorporated in future versions to enhance its performance and usability.

The following enhancements are recommended:

1. Larger Medical Dataset

The current project uses the Heart Disease UCI dataset containing approximately 303 patient records. Future work can utilize larger datasets collected from hospitals and healthcare organizations to improve model accuracy and generalization.

2. Advanced Machine Learning Algorithms

More advanced machine learning algorithms can be implemented, such as:

- Random Forest
- XGBoost
- Support Vector Machine (SVM)
- Gradient Boosting
- Artificial Neural Networks
- Deep Learning Models

These algorithms may improve prediction accuracy for complex datasets.

3. Real-Time Prediction System

The prediction model can be integrated with hospital databases to analyze patient information in real time. Doctors can obtain immediate prediction results during patient consultation.

4. Web-Based Application

The machine learning model can be deployed as a web application using frameworks such as:

- Flask
- Django
- Streamlit

This would allow healthcare professionals to access the prediction system through a web browser.

5. Mobile Healthcare Application

The prediction model can be integrated into Android or iOS mobile applications, enabling doctors and patients to access the system from anywhere.

6. Integration with Electronic Health Records (EHR)

The system can automatically retrieve patient information from hospital Electronic Health Record systems, eliminating manual data entry and reducing errors.

7. Cloud-Based Deployment

The prediction model can be deployed on cloud platforms such as:

- Microsoft Azure
- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)

Cloud deployment would improve scalability and accessibility.

8. Explainable Artificial Intelligence (XAI)

Future systems can incorporate Explainable AI techniques to explain why the model predicts a patient as high-risk or low-risk. This will increase trust and transparency for healthcare professionals.

9. IoT Integration

Wearable health monitoring devices can continuously collect patient health parameters such as:

- Heart Rate
- Blood Pressure
- Oxygen Saturation
- ECG Signals

These data can be sent directly to the prediction model for continuous health monitoring.

10. Multi-Disease Prediction

The proposed system can be expanded to predict multiple diseases using the same platform, including:

- Diabetes
- Kidney Disease
- Liver Disease
- Parkinson's Disease
- Stroke
- Hypertension

This would transform the system into a comprehensive healthcare prediction platform.

7.3 Project Outcomes

The project successfully achieved the following outcomes:

- Developed a machine learning-based heart disease prediction system.
 - Successfully preprocessed the medical dataset.
 - Implemented Logistic Regression and Decision Tree algorithms.
 - Evaluated the models using standard performance metrics.
 - Generated predictions based on patient medical information.
 - Demonstrated the practical application of machine learning in healthcare.
 - Improved understanding of supervised machine learning algorithms and healthcare analytics.
-

7.4 Summary

This chapter presented the overall conclusion of the project and discussed possible future enhancements. The proposed heart disease prediction system demonstrates that machine learning can provide accurate and reliable disease prediction using patient medical records. With larger datasets, advanced algorithms, and deployment as web or mobile applications, the proposed system can evolve into a valuable decision-support tool for healthcare professionals.
